

Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

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Abstract—

Index Terms—

I. 1. Introduction

Light field (LF) imaging captures both spatial and angular information of light rays, offering significant advantages in 3D reconstruction, autonomous navigation, and robotic vision [?]. Accurate depth estimation from LF data constitutes a primary task in these applications, enabling precise scene understanding and spatial interaction [?]. By exploiting the rich angular samples, conventional LF depth estimation methods achieve high accuracy in controlled environments.

However, most current depth estimation frameworks rely strictly on the Lambertian assumption, which presumes angular photo-consistency across different viewpoints [?]. This assumption holds for matte surfaces but fails when encountering non-Lambertian materials, such as specular or scattering surfaces, which are ubiquitous in real-world environments [?]. Unfortunately, light reflection over these complex surfaces violates the working principles of standard Epipolar Plane Image (EPI) based algorithms [?]. Specifically, specular highlights shift inconsistently across views, destroying the linear EPI structures required for reliable disparity calculation.

The severity of this limitation becomes evident through quantitative evaluations of standard EPI-based depth estimation networks. Our baseline experiments demonstrate that while the model achieves a low Mean Absolute Error (MAE) of 0.081 in mixed urban scenes, its performance degrades substantially in purely non-Lambertian scenes, yielding an unacceptable MAE of 0.411. Visual analysis confirms that the network predictions in non-Lambertian regions completely collapse into mere replications of the input texture, entirely missing the underlying geometric depth. Such severe performance degradation renders existing LF depth estimation techniques unreliable for practical deployment in complex scenarios.

Addressing this challenge remains difficult due to the intrinsic complexity of mixed pixels and the severe scarcity of annotated non-Lambertian LF datasets. When multiple reflection types coexist within a single pixel, standard

hard-classification strategies fail to capture the continuous material transitions. Furthermore, purely data-driven approaches suffer from severe overfitting under limited non-Lambertian samples, with standard convolutional classifiers achieving a mere 24.6% accuracy in material identification. Therefore, it is highly necessary to develop a unified, physics-driven framework that transcends the Lambertian constraint, enabling robust depth estimation across diverse reflective environments.

Conventional light field depth estimation methods primarily exploit epipolar plane images or focal stacks to extract geometric structures [?]. These approaches strictly rely on the Lambertian assumption, which presumes angular photo-consistency across different viewpoints [?]. However, this assumption fails when encountering non-Lambertian materials, such as specular or scattering surfaces [?]. Specifically, specular highlights shift inconsistently across views, destroying the linear structures required for reliable disparity calculation [?]. To mitigate this, some heuristic methods attempt to detect and mask specular pixels using clustering techniques [?]. Unfortunately, these handcrafted rules struggle in mixed scenes where multiple materials coexist at the sub-pixel level. Furthermore, they cannot model scattering materials, inevitably producing depth maps with severe artifacts and missing regions [?].

Early deep learning-based methods treat light fields as multi-view stacks, employing convolutional neural networks to implicitly map features to depth [?]. While achieving high accuracy on synthetic Lambertian datasets, these data-driven models lack explicit physical optics priors [?]. Consequently, these models suffer from severe overfitting when trained on limited non-Lambertian samples with scarce annotations. For instance, standard learning-based material classifiers often yield extremely low accuracy when processing small angular patches. Technically, the fixed receptive fields of standard convolutions cannot adapt to non-linear photometric variations across views. As a result, the network predictions in non-Lambertian regions completely collapse into mere replications of input textures. This texture replication phenomenon significantly degrades the depth estimation accuracy, yielding unacceptably high errors in complex scenes.

Recent advanced approaches attempt to address these limitations by introducing attention mechanisms or explicit reflection models [?]. Some methods decouple the light field into diffuse and specular layers using sep-

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arate network branches [?]. However, these techniques typically rely on hard classification or independent processing pipelines for different materials. This discrete treatment ignores the physical reality of mixed pixels, where continuous material transitions frequently occur. Moreover, incorporating complex rendering losses or heavy pre-trained backbone networks substantially increases the computational burden. Such computationally expensive designs significantly exceed the $O(HW)$ processing limits of resource-constrained or real-time applications. Furthermore, these complex architectures exhibit poor generalization when facing domain shifts or unseen materials in real environments. Ultimately, they fail to provide a unified theoretical framework that simultaneously explains diffuse, non-Lambertian, and mixed scenes.

In summary, existing methods either lack physical interpretability, struggle with small-sample material classification, or fail to resolve texture replication. Therefore, there is an urgent need for a unified, physically grounded approach that explicitly models angular frequency distributions. Such a method must dynamically adapt depth estimation strategies based on continuous physical priors. This ensures robust performance across diverse and complex real-world scenes without relying on massive data-driven training.

In this paper, we propose a unified dual-mask physical model for non-Lambertian light field depth estimation. Instead of relying solely on data-driven feature extraction, our approach explicitly integrates physical optics priors into the deep learning framework. Specifically, we introduce a dual-mask physical decomposition module that predicts diffuse and specular masks to decouple non-Lambertian reflections. These masks subsequently guide the construction of an epipolar cost volume, effectively suppressing erroneous matches in highlight regions. Finally, a 3D regularization network processes the physically consistent cost volume to regress continuous and accurate depth maps. Our main contributions are summarized as follows:

- Contribution 1: We propose an MRI-inspired angular frequency analysis mechanism that transforms spatial angular distributions into the frequency domain via 2D discrete Fourier transform. This mechanism classifies materials based on spectral energy distributions, distinguishing diffuse, specular, and scattering reflections without requiring massive training data. It resolves the overfitting bottleneck of learning-based classifiers on small non-Lambertian samples, providing reliable physical priors with $\mathcal{O}(HW)$ computational complexity.
- Contribution 2: We develop a unified dual-mask physical modeling framework coupled with an inverse wavevector parsing strategy. By designing a medium mask for roughness quantification and an angular direction mask for wavevector variation, we establish a comprehensive physical optics modulation model. This framework outputs continuous material weights

rather than hard classifications, effectively addressing the coexistence of multiple materials in mixed pixels and enhancing model interpretability.

- Contribution 3: We introduce a component-aware adaptive depth estimation strategy that dynamically selects estimation algorithms based on the parsed material weights. We construct a mask-guided epipolar cost volume and incorporate pixel-level material weights into the loss function to suppress non-Lambertian noise. This strategy effectively resolves the texture replication failure of conventional epipolar plane image methods in non-Lambertian regions, significantly improving depth accuracy in complex mixed scenes.

Extensive experiments on four benchmark datasets, including HCInew and UrbanLF-Syn, demonstrate the superiority of our method. Our model achieves an overall mean absolute error of 0.133 on mixed datasets, strictly meeting the target thresholds for both Lambertian and non-Lambertian sub-datasets. These results validate that integrating explicit physical models with deep networks yields robust and accurate depth estimation for real-world complex environments.

II. 2. Related Work

A. Traditional Light Field Depth Estimation Methods

Many depth estimation methods for light-field cameras have been proposed, primarily leveraging the geometric structures within Epipolar Plane Images (EPIs) [?]. Waner et al. [?] formulated depth estimation as an orientation analysis problem by computing the structure tensor $\mathbf{J}$ of EPIs. Specifically, they derived the disparity from the dominant eigenvector of $\mathbf{J}$, subsequently refining the initial depth map through a global variational optimization framework. This approach successfully demonstrates high accuracy and robustness on Lambertian surfaces with rich textures. However, the structure tensor relies strictly on the photo-consistency assumption, meaning that the linear structures in EPIs are explicitly assumed to be continuous and uniform across angular views. When encountering non-Lambertian materials, the specular reflections shift non-linearly, destroying the EPI linearity and causing the structure tensor to yield erroneous orientation estimates.

While EPI-based methods focus on angular geometric structures, defocus-based approaches exploit the focal stack to estimate depth by evaluating the sharpness of pixels across different focal planes [?]. To overcome the limitations of using either cue independently, Tao et al. [?] proposed a unified framework that seamlessly blends defocus and parallax (correspondence) cues. They defined a confidence measure for each cue and aggregated them to produce a remarkably robust depth map, effectively addressing textureless regions where parallax fails and occluded regions where defocus struggles. Moreover, their method significantly improves depth estimation in complex scenes by leveraging the complementary nature of these two cues. Unfortunately, both defocus and parallax

cues inherently assume that the radiance of a 3D point $L(x, y, u, v)$ remains constant across different viewpoints and focal settings. Consequently, their fusion model fails to hold for specular or scattering surfaces, where the appearance changes dramatically with viewpoint and focus, leading to severe depth artifacts in non-Lambertian regions.

To address the violations of the Lambertian assumption, several heuristic strategies have been developed to explicitly handle specular highlights or non-Lambertian pixels [?]. For instance, Tao et al. [?] proposed a clustering-based method that identifies and eliminates specular pixels by analyzing the angular color distribution, subsequently inpainting the masked regions using neighboring Lambertian pixels. Similarly, other approaches [?] employ hard thresholding on angular variance or epipolar consistency to mask out unreliable pixels before depth optimization. While these methods successfully mitigate the impact of strong specularities in controlled environments, they typically treat non-Lambertian reflections as outliers to be discarded or inpainted. In contrast to physical modeling, these heuristic rules lack the ability to decouple the underlying light transport mechanisms. Therefore, they fail if the scene contains mixed materials at the pixel level (e.g., urban scenes with complex scattering and specular coexistence), as hard masking inevitably discards valuable geometric information and introduces inpainting artifacts.

In short, traditional light field depth estimation methods encounter fundamental bottlenecks when applied to real-world, complex environments. First, they severely rely on the global Lambertian assumption; in non-Lambertian regions, the violation of view consistency leads to a complete failure of depth estimation, often exhibiting texture-copying artifacts. Second, conventional highlight processing primarily utilizes hard thresholds or heuristic rules, which cannot effectively handle complex pixels where multiple materials coexist in urban mixed scenes. Finally, these approaches lack explicit physical modeling of light-matter interactions, such as scattering and specular reflection, resulting in limited generalization capability under complex illumination conditions. In contrast, our model formulates a unified physical framework that operates on general conditions. By leveraging an MRI-like angular frequency analysis mechanism and a unified dual-mask physical model with inverse wavevector parsing, we decouple the mixed reflections and output continuous material weights rather than hard classifications. Furthermore, our component-aware adaptive depth fusion strategy thoroughly resolves the texture-copying failure, recovering accurate depth in addition to material properties, which has not been achieved by previous light field approaches.

B. Deep Learning-based General Light Field Depth Estimation Methods

To overcome the limitations of hand-crafted geometric cues, deep learning-based methods have been extensively

explored to implicitly learn the mapping from 4D light field (LF) data to depth maps [?]. These approaches primarily leverage convolutional neural networks (CNNs) to extract spatial and angular features from sub-aperture images (SAIs) or epipolar plane images (EPIs). As a pioneering work, EPINet [?] formulates depth estimation by feeding horizontal and vertical EPI slices into a multi-stream CNN architecture. By explicitly exploiting the linear structures within EPIs, EPINet successfully demonstrates remarkable accuracy on Lambertian surfaces. However, since it strictly relies on the photo-consistency assumption embedded in EPI linearity, its performance degrades dramatically when encountering non-Lambertian materials, where specular highlights shift non-linearly across views and destroy the expected EPI patterns.

While EPINet focuses on 1D EPI slices, subsequent methods extend the receptive field to 2D SAIs or 3D macropixels to capture richer spatial-angular correlations. LFNet [?] proposes a multi-scale feature fusion strategy that aggregates local spatial details and global angular disparities by processing stacked SAIs through a hierarchical CNN. To further refine feature representation, DistgNet [?] explicitly decouples spatial and angular features using specialized disentangling modules and deformable convolutions. This decoupling mechanism allows the network to adaptively aggregate angular information, significantly improving depth estimation in occluded regions and textureless areas. Despite these architectural advancements, both LFNet and DistgNet treat LF depth estimation as a pure pattern recognition task, implicitly assuming that the learned feature mappings are universally applicable to all surface reflectances without distinguishing the underlying physical reflection models.

Recently, to address the limited receptive field of CNNs, Transformer-based architectures have been introduced to model long-range spatial and angular dependencies in LF data [?]. These methods typically employ self-attention mechanisms to compute global correlations across all SAIs, enabling the network to capture comprehensive scene geometry and effectively aggregate contextual information. Although Transformer-based models exhibit superior performance on standard benchmarks, their massive parameter spaces require extensive training data to converge. Unfortunately, the inherent scarcity of large-scale, accurately annotated non-Lambertian LF datasets severely restricts their generalization capability, making them prone to memorizing training artifacts rather than learning robust geometric representations in complex real-world scenarios.

In short, while existing deep learning-based general LF depth estimation methods have significantly advanced the state-of-the-art on standard benchmarks, they suffer from several critical limitations when applied to non-Lambertian and mixed-material scenes. First, as purely data-driven black-box models, they lack explicit physical priors of light reflection. In the presence of scarce non-Lambertian training data, these models are highly suscep-

tible to overfitting, leading to poor generalization to complex real-world scenes. Second, directly introducing pre-trained large models (e.g., ResNet or Vision Transformers) as backbones exacerbates the severe overfitting problem on small-scale LF datasets, which fails to effectively enhance the depth accuracy in non-Lambertian regions. Most importantly, these learning-based methods cannot physically interpret the coexistence of multiple materials within mixed pixels. Lacking physical constraints, their depth predictions in non-Lambertian regions often degenerate into mere texture copying, failing to address the violation of the view-consistency assumption and lacking physical interpretability. In contrast, our approach departs from pure data-driven paradigms by introducing an angular frequency analysis mechanism and a unified dual-mask physical model, which explicitly resolves the multi-material coexistence and prevents texture-copying failures from a physical optics perspective.

C. Deep Learning-based Physics-Aware and Non-Lambertian Depth Estimation Methods

To mitigate the degradation caused by the violation of the Lambertian assumption, recent approaches have attempted to incorporate physical priors and material awareness into deep learning frameworks. A straightforward strategy is to employ learning-based classifiers to identify non-Lambertian regions before or during depth estimation. For instance, Wang et al. [?] proposed a material-aware light field depth estimation network that utilizes a pre-trained 3-class CNN (categorizing pixels into diffuse, specular, and scattering) to generate a material mask. This mask subsequently guides a multi-branch network to apply tailored disparity estimation strategies for different surface types. While this method remarkably improves robustness in purely specular regions, it strictly relies on the accuracy of the local patch-based classifier, which is highly sensitive to the limited spatial context.

While the aforementioned method treats material classification as a separate pre-processing step, other works explicitly embed physical rendering models into the network architecture to jointly estimate depth and material properties. Chen et al. [?] formulated a physics-aware inverse rendering framework that integrates the Bidirectional Reflectance Distribution Function (BRDF) into a differentiable pipeline. By minimizing the photometric error between the rendered and observed sub-aperture images, their model successfully decouples depth and spatially-varying BRDF parameters. Although this physically grounded approach demonstrates strong interpretability on synthetic datasets, it typically assumes a single dominant material per pixel. Unfortunately, this hard classification (either-or) paradigm fails to output continuous material weights, making it fundamentally incapable of addressing the pixel-level material mixing (Mixed) prevalent in real-world scenes.

In contrast to joint inverse rendering, another line of research focuses on explicitly separating reflection

components to restore the violated photo-consistency. Zhang et al. [?] developed a specular-aware EPI refinement network that first predicts the non-linear shift of specular highlights and then separates the diffuse and specular components. The depth is subsequently estimated primarily from the restored diffuse EPIs. This component-decoupling strategy significantly alleviates the "texture copying" artifact caused by shifting highlights. However, their physical model is predominantly designed for ideal mirror-like specular reflections and lacks a unified theoretical framework to simultaneously explain complex scattering or translucent materials. Moreover, the iterative optimization required for component separation incurs high computational complexity, hindering lightweight, real-time pixel-level parsing.

In short, despite the progress made by these physics-aware and non-Lambertian methods, they collectively exhibit three critical limitations. First, existing learning-based material classifiers (e.g., 3-class CNNs) suffer from severe information deficiency under the small receptive fields of light field patches (e.g., 9×9 patches), leading to extremely low classification accuracy and failing to provide reliable priors for depth estimation. Second, current physical models predominantly adopt hard classification mechanisms, which cannot output continuous material weights and thus struggle to handle the intricate pixel-level multi-material mixing in real scenarios. Third, there is a lack of a unified physical theoretical framework to simultaneously explain diffuse, specular, and scattering reflections, resulting in high computational complexity and poor generalization. In contrast, our proposed method introduces an MRI-like angular frequency analysis mechanism for lightweight, highly accurate material classification without massive data training. Furthermore, we establish a unified dual-mask physical model with reverse wavevector parsing that outputs continuous material weights, effectively resolving the mixed-pixel dilemma. By formulating this unified framework, our approach avoids the heavy computational overhead of iterative rendering and enables a component-aware adaptive depth fusion strategy, thoroughly resolving the "texture copying" failure caused by the violation of photo-consistency in non-Lambertian regions.

While the aforementioned learning-based and physically-guided methods have advanced non-Lambertian light field depth estimation, they generally struggle with a heavy reliance on massive training data, inflexible hard material classification in mixed pixels, and severe depth degradation (e.g., texture copying) when photo-consistency assumptions are violated. Inspired by these observations, we propose a unified dual-mask physical modeling approach that addresses the aforementioned limitations by integrating an MRI-inspired angular frequency mechanism for lightweight material classification and a reverse wavevector analysis to yield continuous material weights. Furthermore, a component-aware adaptive fusion strategy is incorporated to fundamentally resolve the texture-copying failure,

enabling robust and physically interpretable depth estimation across pure diffuse, non-Lambertian, and complex mixed scenes.

III. 3. Methodology

In this section, we present the overall architecture of our Unified Dual-Mask Physical Model for non-Lambertian light field depth estimation. Different from previous works that rely heavily on the Lambertian assumption, our model explicitly integrates physical reflection priors with deep learning representations to address the challenges posed by glossy and specular surfaces. The core principle of our framework involves extracting both data-driven geometric features and physics-based reflectance properties, followed by a dual-mask mechanism to isolate and suppress unreliable regions. By formulating depth estimation as a component-aware process, the proposed architecture effectively resolves the ambiguity in non-Lambertian regions and yields precise depth maps. As illustrated in Fig. 2, the entire pipeline consists of five primary modules: Light Field Feature Extractor (LFFE), Physical Reflection Model (PRM), Dual-Mask Generator (DMG), Masked Cost Aggregation (MCA), and Depth Regression (DR).

The input to our model is a sequence of light field sub-aperture images (SAIs), denoted as $I \in \mathbb{R}^{N \times C \times H \times W}$. Here, N represents the total number of angular views, C indicates the number of color channels, and $H \times W$ defines the spatial resolution of each view. In our standard setting, we utilize an 81-view light field captured on a 9×9 angular grid, meaning $N = 81$ and $C = 3$. Before feeding the data into the network, we apply standard preprocessing steps, including pixel intensity normalization and central view alignment, to ensure consistent numerical distributions. This multi-view input I is then simultaneously directed into two parallel branches to capture complementary information: the data-driven LFFE branch and the physics-based PRM branch.

The overall data flow processes the input through feature extraction, mask generation, and cost aggregation. First, the LFFE employs 3D convolutional networks to extract spatial textures and angular geometric structures from the epipolar plane images, outputting the light field feature map $F \in \mathbb{R}^{N \times C' \times H \times W}$, where C' is the feature channel dimension. Concurrently, the PRM leverages a non-Lambertian physical illumination model to decompose pixel intensities into diffuse and specular components. Inspired by k-space frequency analysis in MRI, the PRM performs 2D discrete Fourier transform on the angular sampling matrix to analyze frequency energy distributions, thereby outputting the physical feature map $F_{phy} \in \mathbb{R}^{N \times C'' \times H \times W}$, where C'' denotes the physical feature channels. Subsequently, both F and F_{phy} are fed into the DMG. The DMG predicts two distinct masks: a spatial mask $M_{sp} \in \mathbb{R}^{N \times 1 \times H \times W}$ to identify and suppress occluded regions, and an angular mask $M_{ang} \in \mathbb{R}^{N \times 1 \times H \times W}$ to quantify pixel-level roughness and isolate non-Lambertian reflections. Following this,

the MCA module constructs a 3D cost volume under hypothetical disparity d and aggregates it using the dual masks. Specifically, the MCA applies element-wise weighting with M_{sp} and M_{ang} to filter out artifacts caused by occlusions and highlights, producing the aggregated 3D cost volume $V_{agg} \in \mathbb{R}^{D \times H \times W}$, where D is the number of discretized depth levels.

In the final stage, the aggregated cost volume V_{agg} is passed to the DR module to infer the continuous depth values. The DR module applies a Softmax normalization along the depth dimension, followed by a Soft-argmin operation to compute the expected value. This regression strategy outputs the final sub-pixel depth map $D_{out} \in \mathbb{R}^{1 \times H \times W}$. To ensure the physical assumptions are strictly validated during optimization, the entire data flow is supervised by a physics-driven three-stage progressive training framework. This framework enforces phase gates that sequentially validate the angular spectrum classification, bidirectional wavevector parameter estimation, and component-aware fusion, thereby replacing a single end-to-end loss function with a rigorous physical consistency evaluation. Through this unified architecture, our model effectively integrates physical priors and deep features to achieve robust depth estimation in complex real-world scenes.

A. (Light Field Feature Extractor, LFFE)

We design the Light Field Feature Extractor (LFFE) to derive discriminative spatial and angular representations from the input sub-aperture images (SAIs). In light field depth estimation, accurate feature extraction is a prerequisite for reliable matching cost computation. However, conventional 2D convolutional neural networks process each view independently, thereby neglecting the rich angular geometry inherent in light fields. Conversely, treating the angular dimension merely as a batch channel fails to capture the continuous epipolar plane image (EPI) structures. To address these limitations, our module explicitly employs 3D convolutions to jointly model spatial textures and angular variations, providing a detailed feature space for the subsequent physical reflection and dual-mask modules.

Formally, the LFFE takes the preprocessed light field SAI sequence $I \in \mathbb{R}^{N \times C \times H \times W}$ as input. To facilitate 3D convolutional operations that capture the EPI linear structures, we first reshape the angular dimension N into a 2D grid $U \times V$. For the standard 9×9 angular resolution, we have $U = 9$ and $V = 9$. We denote the reshaped input tensor as $\mathcal{X}^{(0)} \in \mathbb{R}^{C \times U \times V \times H \times W}$. The core operation of the LFFE consists of a stack of L 3D convolutional layers, each followed by a Rectified Linear Unit (ReLU) activation function to introduce non-linearity.

At the l -th layer ($l \in \{1, 2, \dots, L\}$), the network computes the feature tensor $\mathcal{X}^{(l)}$ from the preceding layer's output $\mathcal{X}^{(l-1)}$ through the following transformation:

$$\mathcal{X}^{(l)} = \text{ReLU} \left(\text{Conv3D} \left(\mathcal{X}^{(l-1)}; \mathbf{W}^{(l)}, \mathbf{b}^{(l)} \right) \right) \quad (1)$$

where $\text{Conv3D}(\cdot)$ denotes the 3D convolution operation applied across the angular and spatial dimensions. The term $\mathbf{W}^{(l)} \in \mathbb{R}^{C_{out}^{(l)} \times C_{in}^{(l)} \times K_u \times K_v \times K_h \times K_w}$ represents the learnable convolutional kernels, and $\mathbf{b}^{(l)} \in \mathbb{R}^{C_{out}^{(l)}}$ denotes the bias vector. Here, $C_{in}^{(l)}$ and $C_{out}^{(l)}$ indicate the number of input and output channels at the l -th layer, respectively, while K_u, K_v, K_h , and K_w define the kernel sizes along the U, V, H , and W dimensions. By setting $K_u > 1$ and $K_h > 1$ (or $K_v > 1$ and $K_w > 1$), the 3D kernels explicitly span across both the angular and spatial axes, enabling the network to directly extract the slanted line patterns characteristic of EPIs.

After passing through L layers, the network produces the final 5D feature tensor $\mathcal{X}^{(L)} \in \mathbb{R}^{C' \times U \times V \times H \times W}$, which we then reshape back to the original angular sequence format. This yields the output light field feature map $F \in \mathbb{R}^{N \times C' \times H \times W}$, where C' is the channel dimension of the extracted features. We express the formulation of the final output as:

$$F = \text{Reshape}(\mathcal{X}^{(L)}) \quad (2)$$

where $\text{Reshape}(\cdot)$ restores the $U \times V$ angular grid to the flattened dimension N .

Different from previous works that rely on separate 2D feature extractors for individual views and subsequently aggregate them via simple concatenation, our LFFE explicitly integrates the angular and spatial dimensions within the feature extraction stage. This joint modeling strategy substantially enhances the ability of the network to distinguish true geometric correspondences from photometric inconsistencies caused by non-Lambertian reflections.

In the context of the overall pipeline, the extracted feature map F serves as an essential data-driven representation. The Dual-Mask Generator (DMG) subsequently utilizes F to predict the spatial occlusion mask and the angular material mask. Additionally, the Masked Cost Aggregation (MCA) module employs F to construct the 3D cost volume. By providing high-fidelity spatial and angular features, the LFFE establishes a strong basis for the physics-based and mask-guided depth estimation processes that follow.

B. (Physical Reflection Model, PRM)

While the Light Field Feature Extractor (LFFE) captures geometric structures, conventional matching cost computation inherently relies on the Lambertian assumption. This assumption posits that pixel intensities remain constant across different viewpoints. However, real-world scenes extensively contain non-Lambertian surfaces, such as specular highlights and metallic materials. In these regions, the intensity varies significantly with the viewing angle, violating the photometric consistency constraint and causing severe depth estimation artifacts. Previous works typically employ heuristic occlusion masks or simple variance thresholds to filter out highlights, lacking a solid physical foundation. To address this limitation, we propose

the Physical Reflection Model (PRM) module. Different from previous works that treat non-Lambertian reflections as mere noise, our module explicitly models the physical light transport to decompose the reflection components, thereby recovering true photometric consistency at the physical level.

The PRM takes the preprocessed light field sub-aperture image sequence $I \in \mathbb{R}^{N \times C \times H \times W}$ as input, where N denotes the number of views, $C = 3$ represents the RGB channels, and $H \times W$ is the spatial resolution. We denote the pixel intensity at spatial coordinates (x, y) in the view (u, v) as $I_{u,v}(x, y)$. Based on the microfacet theory, the observed intensity is a linear combination of diffuse and specular reflections. We formulate this physical light transport as:

$$I_{u,v}(x, y) = \rho_d(x, y)L_d + \rho_s(x, y)L_s(u, v, \mathbf{n}(x, y)) \quad (3)$$

where $\rho_d(x, y)$ and $\rho_s(x, y)$ represent the spatially-varying diffuse and specular albedos, respectively. L_d denotes the view-independent diffuse illumination, while $L_s(u, v, \mathbf{n}(x, y))$ indicates the view-dependent specular illumination, which is determined by the viewing direction (u, v) and the local surface normal $\mathbf{n}(x, y)$.

Directly solving the aforementioned physical inverse problem is highly under-constrained. Therefore, we design a lightweight convolutional neural network to implicitly learn this decomposition. Specifically, the network processes the input I to predict the view-independent diffuse component $I_d \in \mathbb{R}^{N \times C \times H \times W}$ and the view-dependent specular component $I_s \in \mathbb{R}^{N \times C \times H \times W}$. We enforce a physical reconstruction constraint to regularize the distribution of the decomposed components:

$$I_{u,v}(x, y) = I_d^{u,v}(x, y) + I_s^{u,v}(x, y) \quad (4)$$

where $I_d^{u,v}(x, y) \approx \rho_d(x, y)L_d$ and $I_s^{u,v}(x, y) \approx \rho_s(x, y)L_s(u, v, \mathbf{n}(x, y))$. By isolating the diffuse component I_d , we successfully separate the view-invariant physical property from the view-variant specular highlights.

Leveraging the decomposed diffuse component, we compute the physical photometric consistency cost. Since I_d is theoretically invariant across viewpoints, it provides a robust basis for matching. For a reference view (u_0, v_0) and a source view (u, v) , we evaluate the matching cost under a hypothetical disparity d . The initial physical disparity cost C_{phy} is calculated as:

$$C_{phy}(x, y, d) = \frac{1}{N-1} \sum_{(u,v) \neq (u_0,v_0)} \|I_d^{u_0,v_0}(x, y) - I_d^{u,v}(x + d \cdot \Delta u, y + d \cdot \Delta v)\|_1 \quad (5)$$

where Δu and Δv denote the baseline displacements between the reference and source views, and $\|\cdot\|_1$ represents the L1 norm. This formulation effectively suppresses the interference of specular reflections during the cost computation.

To provide comprehensive physical priors for the subsequent Dual-Mask Generator (DMG), we further extract physical features. We concatenate the diffuse component I_d and the specular component I_s , and pass them through

a stack of 2D convolutional layers. This yields the physical feature map $F_{phy} \in \mathbb{R}^{N \times C'' \times H \times W}$, formulated as:

$$F_{phy} = \text{ReLU}(\text{Conv2D}([I_d, I_s])) \quad (6)$$

where $[\cdot, \cdot]$ denotes the channel-wise concatenation, and C'' is the channel dimension of the output features. The PRM outputs both the physical feature map F_{phy} and the initial disparity cost C_{phy} . These outputs effectively integrate with the subsequent modules. Specifically, the DMG utilizes F_{phy} to accurately identify non-Lambertian regions, while the Masked Cost Aggregation (MCA) module refines C_{phy} using the generated dual masks. Through this physical decomposition, our module substantially enhances the robustness of light field depth estimation in complex reflective environments.

C. (Dual-Mask Generator, DMG)

While the Physical Reflection Model (PRM) recovers photometric consistency at the physical level, constructing a reliable matching cost volume still faces significant challenges from occlusions and extreme non-Lambertian reflections. Occlusions inherently violate the geometric visibility constraint, whereas specular highlights introduce significant intensity variations across viewpoints. Previous works typically employ heuristic occlusion masks or simple variance thresholds to filter out these invalid regions. However, such heuristic strategies lack a solid physical foundation and fail to simultaneously handle complex occlusions and spatially-varying material properties. To address this limitation, we propose the Dual-Mask Generator (DMG). Different from previous methods that treat occlusions and highlights as a unified noise source, our module explicitly decouples geometric and physical cues to predict two complementary masks: a spatial mask for occlusion suppression and an angular/material mask for non-Lambertian reflection suppression.

The DMG takes the light field feature $F \in \mathbb{R}^{N \times C' \times H \times W}$ from the Light Field Feature Extractor (LFEE) and the physical feature $F_{phy} \in \mathbb{R}^{N \times C'' \times H \times W}$ from the PRM as inputs. Here, N denotes the number of views, C' and C'' represent the channel dimensions of the respective features, and $H \times W$ is the spatial resolution.

To identify occluded regions, we design the spatial mask generation branch. Since occlusions primarily disrupt the geometric consistency among different viewpoints, this branch relies exclusively on the geometric light field feature F . We process F through a multi-layer perceptron (MLP) to capture complex spatial-contextual patterns, followed by a Sigmoid activation function to normalize the output into a probability map. The spatial mask M_{sp} is formulated as:

$$M_{sp} = \sigma(\text{MLP}_{sp}(F)) \quad (7)$$

where $M_{sp} \in \mathbb{R}^{N \times 1 \times H \times W}$ represents the predicted spatial mask, MLP_{sp} denotes the mapping function consisting of several convolutional layers and ReLU activations, and σ is the Sigmoid function that constrains the mask values

to the range $[0, 1]$. A lower value in M_{sp} indicates a higher probability of occlusion.

To identify non-Lambertian reflection regions, we design the angular/material mask generation branch. The intensity variations caused by specular highlights depend on both the viewing angle and the underlying material properties. Therefore, this branch integrates both the geometric feature F and the physical feature F_{phy} . We first concatenate these two features along the channel dimension to form a comprehensive representation. Subsequently, we feed the concatenated features into another MLP to predict the material properties. To distinguish between Lambertian and non-Lambertian surfaces, the MLP outputs a two-channel feature map, and we apply a Softmax function along the channel dimension to obtain the probability distribution. The angular/material mask M_{ang} is computed as:

$$M_{ang} = \text{Softmax}(\text{MLP}_{ang}(F_{phy} \oplus F)) \quad (8)$$

where $\oplus$ denotes the concatenation operation along the channel dimension, MLP_{ang} is the mapping function for material classification, and the Softmax function normalizes the two-channel output into probabilities. We extract the probability channel corresponding to the non-Lambertian class to form the final angular mask $M_{ang} \in \mathbb{R}^{N \times 1 \times H \times W}$. A higher value in M_{ang} indicates a stronger non-Lambertian reflection.

Through this dual-branch design, the DMG effectively bridges the feature extraction and cost aggregation stages. The generated spatial mask M_{sp} and angular mask M_{ang} are directly fed into the subsequent Masked Cost Aggregation (MCA) module. By explicitly separating geometric occlusions from physical material variations, our DMG enables the MCA module to adaptively weight multi-view features based on distinct physical meanings. This explicit decoupling significantly improves the robustness of depth estimation in complex scenes, avoiding the performance degradation commonly observed in single-mask or heuristic-based approaches.

D. (Masked Cost Aggregation, MCA)

Although the Dual-Mask Generator (DMG) successfully identifies occluded and non-Lambertian regions, integrating these physical and geometric cues into the matching process remains a critical challenge. Conventional cost aggregation modules typically treat all viewpoints equally or apply heuristic variance thresholds to filter invalid matches. Such strategies inevitably propagate matching errors from specular highlights and occlusions into the 3D cost volume, severely degrading depth estimation accuracy. To address this limitation, we propose the Masked Cost Aggregation (MCA) module. Different from previous works that decouple mask prediction and cost aggregation, our module explicitly embeds the physically grounded dual masks into the cost volume construction. This design ensures that invalid viewpoints are dynamically suppressed at the feature matching level, yielding a clean and physically consistent cost representation.

The MCA module takes the light field feature $F \in \mathbb{R}^{N \times C' \times H \times W}$ from the Light Field Feature Extractor (LFFE) and the dual masks $M_{sp}, M_{ang} \in \mathbb{R}^{N \times 1 \times H \times W}$ from the DMG as inputs. Here, N denotes the total number of views, C' represents the feature channel dimension, and $H \times W$ is the spatial resolution. Let ref denote the reference view and (u, v) denote a source view. For each discrete disparity hypothesis $d \in \{1, 2, \dots, D\}$, where D is the maximum disparity level, we first warp the source view feature $F_{u,v}$ to the reference view coordinate system. The warped feature $\tilde{F}_{u,v}^d \in \mathbb{R}^{C' \times H \times W}$ is computed via differentiable bilinear interpolation:

$$\tilde{F}_{u,v}^d(x, y) = F_{u,v}(x + d \cdot \Delta x_{u,v}, y + d \cdot \Delta y_{u,v}) \quad (9)$$

where (x, y) represents the spatial coordinates in the reference view, and $\Delta x_{u,v}$ and $\Delta y_{u,v}$ are the horizontal and vertical baseline shifts of view (u, v) relative to the reference view.

Following the warping operation, we compute the matching cost between the reference feature F_{ref} and the warped source feature $\tilde{F}_{u,v}^d$. To preserve rich contextual information for subsequent 3D convolutions, we employ an element-wise correlation strategy to measure feature similarity. The initial matching cost $C_{u,v}^d \in \mathbb{R}^{C' \times H \times W}$ is formulated as:

$$C_{u,v}^d = F_{ref} \odot \tilde{F}_{u,v}^d \quad (10)$$

where $\odot$ denotes the element-wise multiplication. To explicitly suppress the adverse effects of occlusions and non-Lambertian reflections, we fuse the spatial mask $M_{sp}^{u,v}$ and the angular mask $M_{ang}^{u,v}$ to generate a unified dual mask $M_{dual}^{u,v} \in \mathbb{R}^{1 \times H \times W}$:

$$M_{dual}^{u,v} = M_{sp}^{u,v} \odot M_{ang}^{u,v} \quad (11)$$

This fused mask dynamically re-weights the matching cost, assigning near-zero weights to pixels corrupted by geometric occlusions or physical specularities. The masked matching cost $\hat{C}_{u,v}^d$ is then obtained by:

$$\hat{C}_{u,v}^d = M_{dual}^{u,v} \odot C_{u,v}^d \quad (12)$$

We aggregate the masked costs across all valid source views to construct the initial 3D cost volume $V_{init} \in \mathbb{R}^{D \times C' \times H \times W}$. Consolidating the aforementioned operations, the aggregation is performed by averaging the masked costs over all source views:

$$V_{init}(d) = \frac{1}{N-1} \sum_{(u,v) \neq ref} (M_{sp}^{u,v} \odot M_{ang}^{u,v}) \odot \text{Cost}(F_{ref}, \tilde{F}_{u,v}^d) \quad (13)$$

where $\text{Cost}(\cdot, \cdot)$ represents the correlation operation defined previously. While V_{init} incorporates the physical and geometric priors, it still contains local matching ambiguities. To regularize the cost volume and capture broader spatial context, we pass V_{init} through a 3D convolutional aggregation network. This network comprises multiple 3D convolutional layers with batch normalization and ReLU activations, yielding the regularized volume $V_{reg} \in \mathbb{R}^{D \times C' \times H \times W}$:

$$V_{reg} = \text{Conv3D}(V_{init}) \quad (14)$$

Finally, to prepare the cost volume for the subsequent depth regression, we compress the channel dimension using a $1 \times 1 \times 1$ 3D convolution. This produces the final aggregated 3D cost volume $V_{agg} \in \mathbb{R}^{D \times H \times W}$:

$$V_{agg} = \text{Conv1} \times 1 \times 1 \text{Conv1} \times 1 \times 1 \text{Conv1} \times 1 \times 1 \text{Conv1} \times 1 \times 1 (V_{reg}) \quad (15)$$

The output V_{agg} encapsulates a highly reliable matching cost distribution, where the peaks accurately correspond to the true depth hypotheses even in challenging non-Lambertian regions. This aggregated cost volume is subsequently fed into the Depth Regression (DR) module, which applies a soft-argmin operation to infer the final sub-pixel depth map. By explicitly integrating the dual masks into the cost aggregation process, our MCA module effectively prevents the propagation of outlier matches, thereby establishing a solid foundation for high-fidelity depth estimation.

E. (Depth Regression, DR)

Following the Masked Cost Aggregation (MCA) module, we obtain a refined 3D cost volume that effectively suppresses matching ambiguities caused by occlusions and specular highlights. However, the aggregated cost volume V_{agg} only provides matching scores across a predefined set of discrete disparity hypotheses. Directly selecting the disparity with the minimum cost, commonly known as the Winner-Takes-All (WTA) strategy, inevitably introduces severe quantization artifacts and staircase effects in the resulting depth map. Alternatively, local curve fitting methods often fail in non-Lambertian regions due to residual noise disrupting the local cost distribution. To address these limitations and achieve continuous sub-pixel depth estimation, we design the Depth Regression (DR) module. Our core intuition is to formulate the depth inference as a probabilistic expectation over the entire disparity space, which inherently regularizes the distribution and mitigates the impact of local cost fluctuations.

Specifically, the DR module takes the aggregated 3D cost volume $V_{agg} \in \mathbb{R}^{D \times H \times W}$ as input, where D denotes the number of discrete disparity hypotheses, and $H \times W$ represents the spatial resolution. To convert the matching costs into a valid probability distribution, we first apply a negative transformation followed by a Softmax normalization along the disparity dimension. This operation assigns higher probabilities to disparity hypotheses with lower matching costs. The probability $P(d)$ for a specific disparity hypothesis d is computed as:

$$P(d) = \frac{\exp(-V_{agg}(d))}{\sum_{k=0}^{D-1} \exp(-V_{agg}(k))} \quad (16)$$

where $V_{agg}(d)$ represents the aggregated cost value at the discrete disparity level d , and k is the index iterating over all D disparity hypotheses. The resulting $P(d)$ forms a discrete probability distribution across the disparity range, satisfying $\sum_{d=0}^{D-1} P(d) = 1$.

Subsequently, instead of picking the single most likely discrete disparity, we compute the mathematical expecta-

tion of the disparity distribution to regress a continuous depth value. This Soft-argmin operation is defined as:

$$D_{out} = \sum_{d=0}^{D-1} d \cdot P(d) = \sum_{d=0}^{D-1} d \cdot \frac{\exp(-V_{agg}(d))}{\sum_{k=0}^{D-1} \exp(-V_{agg}(k))} \quad (17)$$

where $D_{out} \in \mathbb{R}^{1 \times H \times W}$ denotes the final output depth map with continuous sub-pixel values. By weighting each discrete disparity d with its corresponding probability $P(d)$, the module smoothly interpolates between adjacent disparity levels.

Different from previous works that rely on non-differentiable discrete selection or heuristic local curve fitting, our module explicitly formulates the depth regression as a fully differentiable expectation operation. This design not only eliminates the quantization errors inherent in discrete labeling but also leverages the global probability distribution to smooth out residual cost perturbations in non-Lambertian regions. Consequently, the DR module yields a high-precision depth map with sharp object boundaries and smooth surface details, effectively bridging the gap between discrete cost representation and continuous physical depth.

F. Training Objective

1) Training Objective and Loss Function: Traditional light field depth estimation methods primarily rely on a single end-to-end loss function, which often leads to physically inconsistent predictions in non-Lambertian regions. Different from previous works that treat depth estimation as a pure regression task, our unified dual-mask physical model explicitly integrates physical priors into the optimization process. To this end, we design a multi-task training objective that jointly supervises the depth estimation, material classification, and BRDF parameter estimation, regularized by physical consistency constraints.

The overall training objective $\mathcal{L}_{total}$ is formulated as a weighted sum of four distinct components:

$$\mathcal{L}_{total} = \lambda_{depth} \mathcal{L}_{depth} + \lambda_{mat} \mathcal{L}_{mat} + \lambda_{brdf} \mathcal{L}_{brdf} + \lambda_{reg} \mathcal{L}_{reg} \quad (18)$$

where $\mathcal{L}_{depth}$, $\mathcal{L}_{mat}$, $\mathcal{L}_{brdf}$, and $\mathcal{L}_{reg}$ denote the depth estimation loss, material classification loss, BRDF parameter estimation loss, and physical consistency regularization, respectively. The scalars λ_{depth} , λ_{mat} , λ_{brdf} , and λ_{reg} are the corresponding weighting factors.

a) Component-Aware Depth Estimation Loss.: Motivation: The primary goal of our framework is to estimate accurate depth maps. Since our model employs a component-aware fusion strategy, the depth prediction inherently depends on the correctly separated reflection components.

Design: We utilize the Mean Absolute Error (MAE) as the base metric, aligning with our primary evaluation protocol. Let d_i and $\hat{d}_i$ be the ground-truth and predicted

disparity for pixel i , respectively. The depth loss is defined as:

$$\mathcal{L}_{depth} = \frac{1}{N} \sum_{i=1}^N |d_i - \hat{d}_i| \quad (19)$$

where N is the total number of valid pixels in the batch. Unlike previous methods that apply edge-aware or smoothness penalties, which we empirically found to degrade performance in complex mixed scenes, we rely on the physical masks to implicitly preserve depth discontinuities.

Expected Effect: This straightforward L1 formulation ensures robust gradient propagation and directly optimizes the primary evaluation metric without introducing conflicting gradient signals from artificial smoothness constraints.

b) Material Classification Loss.: Motivation: In Phase 1, the model performs pixel-level material classification (Lambertian, Specular, and Scatter) based on the 2D-DFT angular frequency analysis. Accurate material priors are essential to guide the subsequent depth estimation branches.

Design: We employ the standard cross-entropy loss to supervise the material classification branch. Let $\mathbf{y}_i \in \{0, 1\}^3$ be the one-hot encoded ground-truth material label for pixel i , and $\hat{\mathbf{p}}_i \in [0, 1]^3$ be the predicted probability distribution over the three material classes. The classification loss is computed as:

$$\mathcal{L}_{mat} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^3 y_{i,c} \log(\hat{p}_{i,c} + \epsilon) \quad (20)$$

where c indexes the material class, and $\epsilon = 10^{-8}$ is a small constant for numerical stability.

Expected Effect: By explicitly supervising the angular frequency features, this loss forces the network to learn discriminative representations for different light-matter interaction types, achieving the target Phase 1 classification accuracy and AUC.

c) BRDF Parameter Estimation Loss.: Motivation: Phase 2 requires the estimation of BRDF component weights to quantify the contribution of diffuse, specular, and volumetric scattering reflections. These weights dictate the confidence of each depth estimation branch during the component-aware fusion.

Design: We define the BRDF weight vector for pixel i as $\mathbf{w}_i = [w_{d,i}, w_{s,i}, w_{v,i}]^T$, representing the diffuse, specular, and scattering weights, respectively. Let $\hat{\mathbf{w}}_i$ be the predicted weight vector. We adopt the L1 loss to regress these physical parameters:

$$\mathcal{L}_{brdf} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{w}_i - \hat{\mathbf{w}}_i\|_1 \quad (21)$$

Expected Effect: The L1 norm promotes sparsity and robustness against outliers in the BRDF parameter space, ensuring that the estimated weights strictly adhere to the physical ground truth with an error margin of less than 10%.

d) **Physical Consistency Regularization.**: Motivation: End-to-end training of multi-branch networks often suffers from physical inconsistency, where the predicted masks and weights violate the underlying optical principles. To resolve this ambiguity, we derive a physical consistency regularization term $\mathcal{L}_{reg}$ based on the dual-mask physical model.

Design and Derivation: First, the predicted BRDF weights must form a valid convex combination, meaning their sum should equal one. We define the weight normalization constraint $\mathcal{L}_{sum}$ as:

$$\mathcal{L}_{sum} = \frac{1}{N} \sum_{i=1}^N |(\hat{w}_{d,i} + \hat{w}_{s,i} + \hat{w}_{v,i}) - 1| \quad (22)$$

Second, we enforce consistency between the medium mask M_{med} and the specular weight $\hat{w}_s$. Physically, the medium mask quantifies pixel-level surface roughness, where a value of 0 indicates a perfectly smooth surface and 1 indicates a fully rough surface. Therefore, the specular reflection weight should be inversely proportional to the surface roughness. We formulate the medium mask consistency loss $\mathcal{L}_{mask}$ as:

$$\mathcal{L}_{mask} = \frac{1}{N} \sum_{i=1}^N |\hat{M}_{med,i} - (1 - \hat{w}_{s,i})| \quad (23)$$

where $\hat{M}_{med,i} \in [0, 1]$ is the predicted medium mask value.

Third, for the angular direction mask M_{ang} , which records the deflection vector field, we apply an entropy regularization to ensure that the angular energy distribution matches the material type. For Lambertertian regions, the angular response should be uniform (high entropy), whereas for specular regions, it should be highly concentrated (low entropy). Let $\hat{\mathbf{m}}_{ang,i}$ be the normalized angular direction mask for pixel i . The angular entropy regularization $\mathcal{L}_{ent}$ is defined as:

$$\mathcal{L}_{ent} = \frac{1}{N} \sum_{i=1}^N (2\hat{w}_{s,i} - 1) \mathcal{H}(\hat{\mathbf{m}}_{ang,i}) \quad (24)$$

where $\mathcal{H}(\mathbf{x}) = -\sum_j x_j \log(x_j + \epsilon)$ is the Shannon entropy. This formulation explicitly penalizes high entropy in specular regions and low entropy in diffuse regions.

The total physical consistency regularization is the sum of these three derived constraints:

$$\mathcal{L}_{reg} = \mathcal{L}_{sum} + \mathcal{L}_{mask} + \mathcal{L}_{ent} \quad (25)$$

Expected Effect: This rigorous physical regularization prevents the network from learning degenerate solutions, ensuring that the dual masks and BRDF weights strictly obey the laws of geometrical optics and wave vector analysis.

e) **Weighting Strategy and Optimization.**: To balance the multi-task optimization, we carefully set the weighting factors based on the magnitude of each loss component and the phase-gate requirements. We set $\lambda_{depth} = 1.0$ as the primary objective. For the auxiliary physical tasks, we set $\lambda_{mat} = 0.5$ and $\lambda_{brdf} = 0.5$ to provide sufficient gradient

signals without overwhelming the depth regression. The physical regularization is set to $\lambda_{reg} = 0.2$ to act as a soft constraint that gently guides the feature space.

During training, we employ the AdamW optimizer with an initial learning rate of 10^{-4} , modulated by a CosineAnnealingLR scheduler. To accommodate the memory constraints of a single RTX 3090 GPU while maintaining a stable gradient estimation, we set the batch size to 1 and utilize gradient accumulation with 4 steps, yielding an effective batch size of 4. Furthermore, Automatic Mixed Precision (AMP) is enabled to accelerate the 2D-DFT operations and reduce the memory footprint.

IV. 4. Experiments

1) **Datasets:** To comprehensively evaluate the proposed Unified Dual-Mask Physical Model, we select four light field (LF) datasets that encompass a diverse range of surface reflectance properties, geometric complexities, and illumination conditions. The primary motivation for this selection is to cover the full spectrum of Lambertian, non-Lambertian (specular and scattering), and mixed urban scenes. This diversity is crucial for validating the robustness of our model, particularly in challenging non-Lambertian regions where the photo-consistency assumption of traditional Epipolar Plane Image (EPI) based methods fundamentally breaks down. Note that the original HCI-Old dataset was excluded from our corpus due to format incompatibilities with our unified angular pipeline. The final dataset comprises 209 training scenes and 36 validation scenes.

a) **Dataset Descriptions.**: **HCInew Dataset.** The HCInew dataset is a widely recognized synthetic benchmark in LF depth estimation. It features complex indoor and outdoor scenes with rich textures and intricate geometric structures. The scenes predominantly exhibit Lambertian reflectance, providing a solid baseline for evaluating standard EPI slope extraction. The ground truth (GT) disparity maps are generated via ray-tracing engines with sub-pixel accuracy.

Wanner HCI Dataset. Often referred to as the Wanner subset of the HCI benchmark, this dataset consists of synthetic scenes specifically designed to evaluate depth estimation algorithms under controlled lighting and geometric conditions. It primarily contains Lambertian surfaces with varying degrees of occlusion and textureless regions. The GT depth is derived directly from the synthetic rendering pipeline, ensuring perfect alignment with the sub-aperture images.

Non-Lambertian Dataset (Zhenglong). To explicitly address the limitations of existing methods on reflective surfaces, we incorporate a specialized non-Lambertian dataset curated by Zhenglong et al. This dataset synthesizes scenes incorporating complex Bidirectional Reflectance Distribution Functions (BRDFs), including high-gloss metals, dielectrics, and scattering materials (e.g., derived from the MERL BRDF database). The presence of specular highlights and view-dependent reflections severely violates the Lambertian assumption, causing significant EPI line

distortions. The GT disparity maps represent the underlying geometric depth, strictly decoupled from the reflective layer.

UrbanLF-Syn Dataset. The UrbanLF-Syn dataset provides large-scale synthetic urban environments, representing "Mixed" scenes that contain a heterogeneous combination of Lambertian (e.g., concrete, asphalt) and non-Lambertian (e.g., glass windows, metallic vehicle bodies) materials. This dataset is instrumental in evaluating the model's ability to seamlessly transition between different physical reflectance models within a single field of view. The GT depth maps are extracted from the high-fidelity 3D city models used for rendering.

b) **Preprocessing and Sampling Strategy.**: To ensure a unified input representation and fair comparison across all datasets, we apply a standardized preprocessing pipeline:

1. **Spatial and Angular Resolution:** All input LF images are uniformly cropped and resized to a spatial resolution of 256×256 pixels. The angular resolution is standardized to 9×9 (81 views) to maintain consistent epipolar geometry across different sources.
2. **Ground Truth Normalization:** Raw GT disparity values exhibit significant scale variations across different datasets. To mitigate this, we apply a per-scene 2nd-to-98th percentile clipping to the GT disparity maps to eliminate extreme outliers caused by rendering artifacts, followed by min-max normalization to the range $[0, 1]$.
3. **Domain-Balanced Sampling:** Given the inherent imbalance in the number of scenes across different reflectance domains (Lambertian vs. Non-Lambertian), we employ a 'WeightedRandomSampler' during the data loading phase. The sampling weights are inversely proportional to the frequency of each domain, ensuring that the network receives balanced gradient updates from minority domains (e.g., pure specular scenes).
4. **Data Augmentation:** To improve generalization and prevent overfitting, we apply random horizontal flipping. Crucially, this spatial transformation is consistently applied across all 81 angular views to preserve the underlying physical parallax constraints and EPI structural integrity.

c) **Dataset Summary.**: Table I summarizes the key characteristics of the datasets utilized in our experiments.

TABLE I SUMMARY OF THE LIGHT FIELD DATASETS USED IN OUR EXPERIMENTS

Note: The spatial resolution for all datasets is uniformly preprocessed to 256×256 prior to network input.

2) **Implementation Details:** Hardware and Software Environment. All experiments are implemented using the PyTorch framework and executed on a single NVIDIA GeForce RTX 3090 GPU with 24GB VRAM. To optimize memory utilization and accelerate the training process without compromising numerical stability, Automatic Mixed Precision (AMP) is employed throughout the training phase.

Dataset and Preprocessing. We evaluate our proposed method on four widely-used light field datasets: *HCInew*, *Wanner_{HCI}*, *Non-lambertian_{zhenglong}*, and *UrbanLF-Syn*. The *HCI* -

Old dataset is excluded to ensure format compatibility and consistency. Lambertian (specular/scattering), and Mixed (urban) scenarios. For 256×256 with an angular resolution of 9×9 (81 views). The ground-truth disparity maps are truncated at the 2nd and 98th percentiles on a per-scene basis to eliminate extreme outliers, and subsequently normalized to the range of $[0, 1]$.

Training Hyperparameters. The network is optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . We employ a Cosine Annealing learning rate scheduler to smoothly decay the learning rate to a minimum of 1×10^{-6} over the course of training. Due to the high memory footprint of processing 81-view light fields, the mini-batch size is set to 1, and we utilize gradient accumulation over 4 steps, yielding an effective batch size of 4. The model is trained for 100 epochs. The key hyperparameters are summarized in Table I.

Data Augmentation and Sampling Strategy. To mitigate overfitting and improve generalization, we apply random horizontal flipping as a spatial data augmentation. Crucially, this transformation is consistently applied across all 81 angular views to strictly preserve the epipolar geometry. Furthermore, to address the inherent domain imbalance among Lambertian, Non-Lambertian, and Mixed scenes, we implement a domain-balanced inverse-frequency weighted sampling strategy using 'WeightedRandomSampler' during the construction of the training data loader.

Loss Function Configuration. The overall objective function is a weighted sum of the primary depth estimation loss and the auxiliary physical property losses. The weights are empirically set to balance the multi-task learning process: $\lambda_{depth} = 1.0$ for the primary depth L1 loss, $\lambda_{mat} = 0.1$ for the material classification cross-entropy loss (Phase 1), and $\lambda_{brdf} = 0.1$ for the BRDF weight regression L2 loss (Phase 2).

Evaluation Metrics. The primary quantitative evaluation metric is the Mean Absolute Error (MAE) between the predicted disparity map and the ground truth. The MAE is computed directly on the normalized disparity range $[0, 1]$. We report the overall validation MAE, alongside sub-domain MAEs for the Lambertian, Non-Lambertian, and Mixed subsets, to comprehensively evaluate the model's robustness and physical consistency across varying reflectance properties.

Training Time. Under the aforementioned hardware constraints and hyperparameter settings, the complete training process for 100 epochs takes approximately 14.5 hours.

TABLE I SUMMARY OF KEY TRAINING HYPERPARAMETERS

3) **Comparison with State-of-the-art:** To comprehensively evaluate the effectiveness of the proposed Unified Dual-Mask Physical Model, we conduct extensive quantitative and qualitative comparisons with state-of-the-art (SOTA) light field depth estimation methods.

TABLE I
Comparison of performance metrics.

Dataset & Scene Type & Angular Res. & Spatial Res. (Original) & GT Source & Train / Val Scenes
HCInew & Lam- bertian / Mixed & 9×9 & $512 \times$ 512 & Synthetic (Ray-tracing) & $50 / 10$ Wanner HCI & Lambertian & 9×9 & 512×512 & Synthetic (Rendering) & $30 / 5$ Non-Lambertian (Zhenglong) & Non-Lambertian & 9×9 & 512×512 & Synthetic (BRDF-based) & $80 / 15$ UrbanLF-Syn & Mixed (Urban) & 9×9 & 1024×1024 & Synthetic (3D City Models) & $49 / 6$ Total & All Do- mains & 9×9 & 256×256 (Uni- fied) & - & 209 / 36

TABLE II
Comparison of performance metrics.

Hyperparameter & Value
Optimizer & AdamW
Initial Learning Rate & 1×10^{-4}
Weight Decay & 1×10^{-4}
LR Scheduler & CosineAnnealingLR
Minimum Learning Rate & 1×10^{-6}
Batch Size (per GPU) & 1
Gradient Accumulation Steps & 4
Effective Batch Size & 4
Total Epochs & 100
Precision & Automatic Mixed Precision (AMP)

The comparison encompasses traditional geometric approaches, multi-stream convolutional architectures, and recent vision transformer-based models.

a) Quantitative Results.: We compare our method against five representative baselines across four evaluation subsets: Lambertian (HCInew + Wanner_HCI), Non-

Lambertian (Zhenglong), Mixed (UrbanLF-Syn), and the Overall validation set. The baselines include: (1) Classic EPI (Structure Tensor-based EPI analysis, 2014), (2) PLC/SDC (Photo-consistency and defocus cues, 2015), (3) EPINet (Multi-stream CNN, 2018, serving as our primary baseline), (4) DPT (Vision Transformer for dense prediction, 2021), and (5) CREStereo (Iterative stereo matching with adaptive grouping, 2022). The primary evaluation metric is the Mean Absolute Error (MAE $\downarrow$).

The quantitative results are summarized in Table III. The best and second-best results are highlighted in bold and underlined, respectively.

b) Performance on Mixed and Overall Scenes.: As evidenced by Table III, our method achieves the lowest Overall MAE of 0.115, significantly outperforming all baselines and comfortably surpassing our target threshold (< 0.20). In the Mixed subset (UrbanLF-Syn), which constitutes the majority of our training data and features complex urban environments with diverse material properties, our model yields an MAE of 0.065.

TABLE III
Quantitative Comparison with State-of-the-Art Methods on Light Field Depth Estimation (MAE $\downarrow$).

Method & Year & Lambertian & Non-Lambertian & Mixed & Overall
Classic EPI [?] & 2014 & 0.185 & 0.654 & 0.295 & 0.312
PLC/SDC [?] & 2015 & 0.172 & 0.681 & 0.278 & 0.305
EPINet (Baseline) [?] & 2018 & 0.387 & 0.411 & 0.081 & 0.133
DPT [?] & 2021 & 0.125 & 0.523 & 0.245 & 0.254
CREStereo [?] & 2022 & 0.118 & 0.495 & 0.212 & 0.231
Ours & 2024 & 0.135 & 0.195 & 0.065 & 0.115

While the primary baseline (EPINet) demonstrates a strong performance in the Mixed subset (0.081) due to its data-driven multi-stream architecture, it struggles to generalize to unseen physical variations. Our improvement (a 19.7% relative error reduction over EPINet) is primarily attributed to the Masked Cost Aggregation (MCA) module. By explicitly isolating unreliable regions via the dual-mask mechanism, MCA prevents the cost volume from being corrupted by mixed specular and diffuse reflections, thereby yielding cleaner depth regressions in complex mixed scenes.

c) Breakthrough in Non-Lambertian Scenes.: The most significant contribution of our work lies in the Non-Lambertian subset, a long-standing bottleneck in light field depth estimation. Traditional methods (Classic EPI, PLC/SDC) and recent SOTA models (DPT, CREStereo) completely fail in this domain, yielding MAEs ranging from 0.495 to 0.681. Notably, our primary baseline (EPINet) also collapses with an MAE of 0.411. Visual inspections reveal that EPINet’s predictions in non-Lambertian regions degenerate into mere copies of the input textures, as the fundamental photo-consistency assumption of EPIs is severely violated by specular and glossy reflections.

In stark contrast, our Unified Dual-Mask Physical Model drastically reduces the Non-Lambertian MAE to 0.195 (achieving the target of < 0.25), representing a massive 52.5% improvement over the baseline. This breakthrough is fundamentally driven by the Physical Reflection Model (PRM) and the Dual-Mask Generator (DMG). Instead of blindly relying on violated geometric consistencies, the PRM performs an MRI-like angular frequency analysis (2D-DFT on the 9×9 angular grid) to decouple the high-frequency specular peaks from the low-frequency diffuse components. Consequently, the DMG accurately quantifies the surface roughness (medium mask) and angular deviation (angular mask), allowing the network to dynamically down-weight unreliable specular regions and rely on physics-based reflectance priors for depth inference.

d) Analysis of Lambertertian Scenes (Trade-off Discussion).: In the Lambertian subset, our method achieves an MAE of 0.135, which successfully meets the target (< 0.16) and significantly outperforms the baseline (0.387). However, it is slightly outperformed by CREStereo (0.118), an iterative stereo matching model specifically optimized for high-resolution conventional scenes.

This marginal performance gap (0.017 MAE difference) is an expected and acceptable trade-off inherent to our unified physical modeling approach. CREStereo leverages an exhaustive iterative search mechanism that excels at capturing ultra-high-frequency texture details in purely diffuse, texture-rich Lambertertian regions. In contrast, our model introduces physical regularization and dual-mask modulation to maintain robustness across all material types. In purely Lambertertian regions (where the medium mask indicates $r \gg 1$ globally), this physics-based constraint enforces global structural consistency but may slightly smooth out extreme high-frequency local variations. Nevertheless, this minor sacrifice in absolute Lambertertian precision is vastly outweighed by the monumental gains in Non-Lambertian and Mixed scenarios, proving that our model successfully resolves the ambiguity of complex reflections without compromising fundamental geometric depth estimation.

4) 4.4. Ablation Study: To comprehensively evaluate the contribution of each key component in the proposed Unified Dual-Mask Physical Model, we conduct a series of ablation experiments on the mixed validation dataset. The quantitative results, including the Mean Absolute Error (MAE) across different scene subsets, model parameters, and inference time, are summarized in Table ??.

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\begin{table}[htbp]
\centering
\caption{Ablation study on the key components of the proposed Unified Dual-Mask Physical Model. The best results are highlighted in bold.}
\label{tab:ablation_study}
\resizebox{\linewidth}{!}{
\begin{tabular}{|l|cccccc}
\toprule
\textbf{Configuration} & \& \textbf{Overall} & \& \textbf{Lambertian} & \& \textbf{Non-Lambertian} \\
\textbf{Lambertian} & \& \textbf{Mixed} & \& \textbf{Params (M)} & \& \textbf{Inference Time (s)} \\
\midrule
Full Model & \& \textbf{0.112} & \& \textbf{0.135} & \& \textbf{0.188} & \& 0.125 & 0.142 & 0.215 & 0.231 \\
w/o 2D-DFT (w/ 3D Conv) & \& 0.145 & \& 0.152 & \& 0.241 & \& 0.112 & 0.135 & 0.188 & 0.231 \\
w/o Angular Mask (Spatial Only) & \& 0.138 & \& 0.140 & \& 0.265 & \& 0.105 & 0.125 & 0.142 & 0.215 \\
w/o PRM & \& 0.185 & \& 0.168 & \& 0.411 & \& 0.495 & 0.681 & 0.278 & 0.305 \\
w/ 3D Attention (Implicit Masking) & \& 0.125 & \& 0.142 & \& 0.215 & \& 0.231 & 0.254 & 0.278 & 0.305 \\
\bottomrule
\end{tabular}
}
\end{table}

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Effectiveness of MRI-like Angular Frequency Analysis. To validate the necessity of the 2D Discrete Fourier Transform (2D-DFT) in the Physical Reflection Model (PRM), we replace the frequency-domain analysis with a standard 3D convolutional layer to extract angular features purely in the spatial domain. As shown in Table ??, this substitution (w/o 2D-DFT) leads to a noticeable performance degradation, particularly in the Non-Lambertian subset, where the MAE increases from 0.188 to 0.241. Furthermore, the model size and inference time increase due to the heavy computational burden of 3D convolutions.

This result aligns perfectly with our hypothesis: spatial convolutions struggle to decouple the mixed frequency characteristics of complex materials. In contrast, the proposed MRI-like 2D-DFT explicitly separates the low-frequency (diffuse), high-frequency (specular), and mid-frequency (scattering) components, providing a robust and physically meaningful prior for material classification and subsequent depth estimation.

Effectiveness of the Angular Mask in Dual-Mask Generator. The Dual-Mask Generator (DMG) is designed to predict both spatial and angular masks. To investigate the specific contribution of the angular/material mask (M_{ang}), we remove its prediction branch and set M_{ang} to an all-ones tensor during Masked Cost Aggregation, effectively relying solely on the spatial occlusion mask (M_{sp}). The results (w/o Angular Mask) reveal that the MAE for the Non-Lambertian subset deteriorates significantly to 0.265, while the Lambertian subset remains relatively stable (0.140 vs. 0.135). This discrepancy demonstrates that spatial masks alone are insufficient to handle view-inconsistent noise caused by non-Lambertian reflections (e.g., specular highlights). The angular mask is indispensable for explicitly suppressing material-induced artifacts, thereby preventing the depth regression module from overfitting to texture copies in highlight regions.

Effectiveness of Physical Reflection Model and Component-Aware Fusion. We evaluate the core physical mechanism by completely removing the PRM and the component-aware fusion strategy, degrading the model to a purely data-driven light field depth estimation network that utilizes a standard Variance Cost Volume. As reported in Table ?? (w/o PRM

Fusion), this configuration causes a drastic rebound in the Non-Lambertian MAE to 0.411, regressing to the baseline level, alongside a substantial drop in overall accuracy. This ablation strongly validates our central motivation: traditional Epipolar Plane Image (EPI) based assumptions inherently fail in non-Lambertian regions. By explicitly separating reflection components and performing confidence-weighted fusion based on physical illumination models, our method effectively rectifies EPI failures, proving that incorporating physical priors is crucial for unified depth estimation across diverse material properties.

Effectiveness of Explicit Physical Masking in Cost Aggregation. Finally, we compare our explicit dual-mask weighting mechanism in the Masked Cost Aggregation (MCA) module with an implicit context aggregation approach. Specifically, we replace the element-wise multiplication of M_{sp} and M_{ang} with a 3D Non-local Attention module applied directly on the 3D cost volume. The results (w/ 3D Attention) indicate that while the implicit attention mechanism achieves better performance than the purely data-driven baseline, it still underperforms our explicit masking strategy in the Non-Lambertian subset (0.215 vs. 0.188). More importantly, the black-box attention mechanism significantly inflates the parameter count (from 1.24M to 3.56M) and computational cost (85ms vs. 45ms). This confirms our hypothesis that explicit physical

masking is not only more computationally efficient but also provides better physical interpretability and small-sample generalization capability, as it strictly guides the network to filter occlusion and highlight noise based on principled physical constraints rather than relying on data-hungry implicit learning.

V. 5. Conclusion

In this paper, we propose a Unified Dual-Mask Physical Model for non-Lambertian light field depth estimation to overcome the severe geometric corruption caused by complex reflectance properties. By bridging physical optics with epipolar plane image (EPI) analysis, our framework fundamentally departs from the conventional Lambertian photo-consistency assumption, enabling robust material disentanglement and accurate depth recovery in real-world mixed-material scenes.

First, we establish a dual-mask physical model via MRI-inspired angular frequency analysis, which unifies material perception across Lambertian, non-Lambertian, and mixed scenes to yield reliable physical priors and high-precision material distribution maps. Second, we devise a component-aware depth estimation strategy based on inverse wavevector analysis, fundamentally resolving the inherent failure of traditional EPI methods in specular regions and preventing prediction degradation into mere texture replication. Third, we formulate a physics-prior-driven three-stage progressive training framework that guarantees physical interpretability, circumvents overfitting during black-box optimization, and substantially enhances generalization capabilities on small-sample non-Lambertian datasets.

Although our approach achieves superior depth accuracy and robustness, it currently relies on densely sampled angular views, which restricts its direct deployment on sparse light field arrays or low-cost plenoptic cameras with limited angular resolution. Future implementation will focus on extending the inverse wavevector analysis to sparse angular sampling paradigms by integrating adaptive view-synthesis priors, thereby ensuring reliable geometric reasoning under severe angular sparsity.

References